Mukesh Patel School of Technology Management & Engineering
School of Technology Management and Engineering

Course Policy Document

Course Name–Basic Electrical and Electronics Engineering (702EX0C001)

<table>
<tr><td>Program and Semester:B.Tech./MBATech. (all branches) Semester-I/II</td><td>Pre-requisiteCourse: Physics and Mathematics (Pre-university level)</td></tr>
<tr><td>Academic Year:2024-25</td><td>Credit Details:
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<tr><td>Name of Course Faculty:
Dr. Ushma Ahuja
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Office hours: 9 am to 4 pm</td><td>Faculty associated with the course: -
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2. Dr. Anjana Rodrigues (Mumbai)
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4. Dr. Vikas Khare (Indore)
5. Dr. Rahul Koshti (Hyderabad)
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10. Prof. Tazeen Shaikh (Mumbai)
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<tr><td colspan="2">Pre-Course Activity: Nil</td></tr>
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1. Introduction to the Course

1.1 Importance of the Course

1.1.1 Domain Relevance:

This course helps students to understand fundamental principles of Electrical Engineering, also it provides an overview of evolution of electronics, and introduces the working principle and examples of electronic devices and circuits to make students understand the impact of technology in a global and societal context.

1.1.2 Industry Relevance:

Electrical and electronics engineering is at the forefront of developing new technologies for several industries including transport, healthcare, construction, and robotics. With the rapid expansion of the technology sector, the demand for those who can create, understand, and alter electrical control systems has risen. Electrical and electronic engineers create and improve systems to benefit society. Electrical

design can become a very complex topic, especially where large installations are concerned (such as energy grids) or even within highly advanced pieces of small hardware (such as those used in smartphones). Therefore, if to be able to work upon such systems, a solid understanding of basic circuit design is required which is offered by this course.

1.2 Objectives of the Course:

1.2.1 To equip the students with the ability to solve, assemble and test simple DC electrical circuits
1.2.2 To equip the students with the ability to solve, assemble and test simple AC electrical circuits
1.2.3 To obtain a basic understanding of the working principle and applications of electronics devices

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

After successful completion of the course, a student will be able to

2.1.1 **CO1:** Interpret DC circuits, theorems and time domain analysis of first order RL circuit

2.1.2 **CO2:** Analyze series and parallel AC circuits and compare star/delta configurations

2.1.3 **CO3:** Comprehend the principles of transformer and electrical machines

2.1.4 **CO4:** Review the construction, working principle and applications of electronics devices and logic circuits

2.2 Program Outcomes(PO) the course contributed to:

2.2.1 **PO-1:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2.2.2 **PO-2:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

2.2.3 **PO-3:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations.

2.2.4 **PO-4:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

2.2.5 PO-5: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

2.2.6 PO-6: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

2.2.7 PO-7: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

2.2.8 PO-8: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

2.2.9 PO-9: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

2.2.10 PO-10: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

2.2.11 PO-11: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

2.2.12 PO-12: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

2.3 Program Specific Outcomes (PSO):

2.3.1 PSO1: Apply the basic sciences and engineering knowledge in the design and development of complex systems in the areas related to electronics and communication engineering

2.3.2 PSO2: Use the cutting-edge hardware and software tools with the obtained technical and managerial skills to design software and systems for applications including signal processing, communication engineering, computer networks, VLSI design, and embedded systems

2.3.3 PSO3: Possess the attitude of continuous learning for producing effective solutions for the applications, directly and indirectly, related to Electronics and Communication engineering

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2			2						1	3		2
CO2	3	3				2						1	3		2
CO3	3					2						1	3		2
CO4	3	2	2			2						1	3		3

Mapping Levels: 1- High, 2-Medium, 3-Low

2.4 Student Outcomes (SO) (For ABET accredited Programs):

2.4.1 SO-1: Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

2.4.2 SO-2: Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

2.4.3 SO-3: Communicate effectively with a range of audiences.

2.4.4 SO-4: Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

2.4.5 SO-5: Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

2.4.6 SO-6: Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

2.4.7 SO-7: Acquire and apply new knowledge as needed, using appropriate learning strategies.

3. Teaching-learning methodology

3.1 Instruction Plan

Lecture No.	Topic	Teaching Method*	Blooms Level	Resources	COs mapped	Assessment and Evaluation
Unit 1: DC Circuits						
1	Electrical Circuit Elements (R, L, C)- Properties of elements, applications. Voltage and Current sources-	Whiteboard and PPT	Understand	TB1, Ch-1, pp. 1.2	CO1	Lab submission, Class Test-1, TEE

	Symbols, polarity. Source transformation					
2	Analysis of simple circuits - Resistance in series and parallel (Theory and numerical). Voltage and Current divider rule and its numerical.	Whiteboard and PPT	Understand	TB1, Ch-1, pp. 1.10 	CO1	Lab submission, Class Test-1, TEE
3	KCL and KVL Statements, Mesh analysis- Steps followed and numerical (Excluding supermesh)	Whiteboard and PPT	Analyse	TB1, Ch-1, pp. 2.2 	CO1	Lab submission, Class Test-1, TEE
4	Nodal analysis- Steps followed and numerical (Excluding supernode)	Whiteboard and PPT	Apply	TB1, Ch-1, pp. 2.48 	CO1	Lab submission, Class Test-1, TEE
5	Superposition Theorem- Statement, Explanation and Numerical. (considering independent voltage and current sources)	Whiteboard and PPT	Apply	TB1, Ch-1, pp. 2.116 	CO1	Lab submission, Class Test-1, TEE
6	Thevenin's Theorem- Statement, Explanation and Numerical.	Whiteboard and PPT	Apply	TB1, Ch-2, pp. 2.153	CO1	Lab submission, Class Test-1, TEE
7	Norton's Theorem- Statement, Explanation and Numerical.	Whiteboard and PPT	Apply	TB1, Ch-2, pp. 2.188	CO1	Lab submission, Class Test-1, TEE
8	Time domain Analysis of first order RL Circuits Derivation for growth of current and decay of current	Whiteboard and PPT	Understand		CO1	Lab submission, Class Test-1, TEE
Unit 2: AC circuits						
9	Generation of alternating voltage, Basic terms related to Alternating quantities (Waveform, cycle,	Whiteboard and PPT	Understand	TB1, Ch-3, pp. 3.2	CO2	Lab submission, Class Test-1, TEE

	frequency, Phase, Phase difference) (Numerical based on instantaneous values, frequency and time period) Root Mean Square, Average value (for sine wave), Mathematical representation of phasors (representation in 2 forms-rectangular form, polar form)					
10	Class Test 1					
11	Analysis of AC circuits - containing resistance only, Inductance only, Capacitance only), Derivation includes parameters such as voltage, current, Waveforms, Phasor diagram, Impedance, phase difference, power factor, and power	Whiteboard and PPT	Understand, Analyse	TB1, Ch-4, pp. 4.2	CO2	Lab submission, Class Test-2, TEE
12	A.C analysis of series R-L circuit, Derivation includes parameters such as current, waveforms, Phasor diagram, impedance, power and comment on power factor.Numerical based on series R-L circuit	Whiteboard and PPT	Understand, Analyse	TB1, Ch-4, pp. 4.10	CO2	Lab submission, Class Test-2, TEE
13	A.C analysis of series R-C circuit, Derivation includes parameters such as current, waveforms, Phasor diagram, impedance, power and comment on power factor, Analysis of series RLC circuit considering 3 cases: (XL>XC), (XL<XC) and (XL=XC)	Whiteboard and PPT	Understand, Analyse	TB1, Ch-4, pp. 4.36	CO2	Lab submission, Class Test-2, TEE
14	Resonance concept, important properties of	Whiteboard and PPT	Understand, Analyse	TB1, Ch-4, 4.90	CO2	Lab submission,

	series resonance circuit, expression of resonant frequency, parameter with respect to the series RLC (Power factor, current, voltage, phasor diagram, expression of quality factor, bandwidth concept, behavior of Behaviors of R, L, C with change in frequency) *(To be covered in lab with graph verification)* Numerical based on series resonance circuit					Class Test-2, TEE
15	Basic introduction to parallel resonance circuit, Polyphase system, Advantages of Three phase system over single phase system, star and delta connections, Relation between phase & line voltages, and phase & line currents in star connections *(no derivation)*	Whiteboard and PPT	Understand, Analyse	TB1, Ch-4, pp. 4.106, Ch-5, pp. 5.2	CO2	Lab submission, Class Test-2, TEE
16	Relation between phase & line voltages and phase & line currents in Delta connections. *(no derivation),* Numerical based on star and delta connections	Whiteboard and PPT	Understand, Analyse	TB1, Ch-5, pp. 5.6	CO2	Lab submission, Class Test-2, TEE
Unit 3: Transformers and Motors						
17	Transformer- construction, working, EMF Equation of 1-phase transformer, Transformation ratio, Full load current calculation. Introduction to voltage regulation and maximum efficiency, Numerical on above topics.	Whiteboard and PPT	Understand	TB1, Ch-6, pp. 6.2	CO3	Class Test-2, TEE

18	Ideal and practical transformer (brief idea, no phasor diagrams and its explanation), Losses in transformer- classification, Equivalent circuit of a transformer (no numerical).	Whiteboard and PPT	Understand	TB1, Ch-6, pp. 6.13	CO3	Class Test-2, TEE
19	DC motor- Construction, working principle, applications, Classification of DC Motor- Shunt and series (only brief introduction with diagram).	Whiteboard and PPT	Understand	TB1, Ch-7, pp. 7.2 	CO3	Lab submission, Class Test-2, TEE
20	3 phase Induction motor- Construction, working principle, and applications.	Whiteboard and PPT	Understand	TB2, Ch-15, pp. 589 	CO3	Class Test-2, TEE
21	Single phase Induction motor- Construction, working principle, and applications.	Whiteboard and PPT	Understand	TB2, Ch-15, pp. 589 	CO3	Class Test-2, TEE
22	**Class Test 2**					
Unit 4: Analog Electronics						
23	Basics of diodes, Diode Rectifiers- Half wave and Full wave: ● Circuit diagram, working and waveforms Zener Diode ● Basic operation ● Application as voltage regulator. BJT (N-P-N) Symbol, working, application as an inverting switch.	Whiteboard and PPT	Apply	TB3, Ch-19, pp. 413	CO4	Lab submission, TEE
24	Opto-electronic devices- Concept, working principle, types of optoelectronics devices (Photo diode, solar cells,	PPT	Apply		CO4	TEE

	LEDs), Opto-coupler- working, and Applications. Transducer- Need, Definition, Block diagram (sensing and transduction element), applications.					
25	Operational amplifier - Block diagram, Inverting and non- inverting amplifier, Op-amp as a comparator and RC phase shift oscillator	Whiteboard and PPT	Apply	TB3, Ch-23, pp.480	CO4	TEE
26	IC-555 timer- Pin diagram and working. Monostable Multivibrators	PPT	Understand		CO4	TEE
Unit 5: Digital Electronics						
27	Logic gates – AND, OR, NOT, Truth Table, Gate Symbol, DeMorgan Theorem Universal gates – NAND, NORTruth Table, Gate Symbol Exclusive gates – XOR, XNOR, Truth Table, Gate Symbol	Whiteboard and PPT	Apply	TB3, Ch-26, pp. 524	CO4	Lab submission, TEE
28	Implementation of Boolean expressions using logic gates with examples (Only using basic logic gates) ● Boolean Laws ● Simplification of sum of products (SOP) and Demonstration of product of sums (POS) function	Whiteboard and PPT	Apply	TB3, Ch-26, pp. 524	CO4	TEE
29	Applications of digital circuits- Adder – Half and Full Adder circuit, Multiplexer – many to one multiplexing, 2:1, 4:1, and	Whiteboard and PPT	Apply	TB3, Ch-26, pp. 543	CO4	TEE

	8:1 data selector schematic demonstration, De-multiplexer – concept of one to many de-multiplexing, schematic demonstration.					
30	Analog to digital Converter, Digital to analog converter	Whiteboard and PPT	Apply	TB3, Ch-24, pp. 591	CO4	TEE

4. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 Marks)				
	Class Tests (20 Marks)		**Term-work (30 Marks)**		
Assessment Component	Class Test-1	Class Test-2	Lab Submissions	Assignments (In-class and Home)	Challenging Problems
Marks	10	10	20	5	5

4.1 Internal Continuous Assessment (ICA) – 50 marks

Assessment of ICA comprises the following components.

1. Class test 1 and 2

a. Two class tests will be conducted as per the academic calendar.
b. It may be conducted for 20 marks each and scaled down to 10 marks each.

2. Lab Submissions (15 marks)

a. Lab experiments (5 marks)

i. Continuous assessment for laboratory experiments will be conducted. **There are 10 practicals, each carrying weightage of 10 marks.** At the end of the course, average of total marks will be taken to obtain marks out of 10 and then scaled down to 5.
ii. Discussion of your work with your peers is allowed. However, each student is expected to submit his/her original work. Submissions which are very similar will be marked zero. Assessment of the lab work will be carried out based on parameters like timely completion of lab work file, understanding of the experiment performed, originality in the work, involvement of the student, regularity, discipline etc. during the session.

The rubrics for evaluation of each experiment is as follows:

Parameter	Involvement	General Conduct	Viva	On time submission	Lab Manual	Total	Scaled down
Total Marks	2	1	2	2	3	10	5
Marks allotted							

b. **Lab test and viva (15 marks)-**Lab test will be conducted based on the concepts learnt. Viva will be based on the syllabus topic. Lab test will be of 10 marks and viva will be of 5 marks.

3. **Assignment (5 marks)-**Tutorials will be given to the students consisting of mostly numerical problems for home assignment or classwork submission.
4. **Implementation of a challenging problem (group activity) (5 marks)**

   Faculty will make group of 2-3 students.

   i. Identify a project based on some topic relevant to the course
   ii. Course faculty will approval the project topic
   iii. Request students to implement it on any software platform (Multisim, Tinker Cad etc.)
   iv. Assessment will be based on the methodology used to implement and understanding of the project.

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

A written examination of 100 marks for three hours' duration will be conducted for the course as per the academic calendar.
Marks obtained scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.3.3 (ICA + TEE) (100 marks) - 40% required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)					Term End Examination (TEE)
Course Outcomes	CT-1	CT-2	Lab Submissions	Assignments	Challenging Problems	TEE
CO-1	Y		Y	Y	Y	Y
CO-2		Y	Y	Y	Y	Y
CO-3			Y	Y		Y
CO-4			Y	Y		Y

5. Laboratory details

The following 10 programming exercises will form the submission for laboratory coursework. Each programming exercise will contain 3 to 5 programs.

Exp. No.	Week No.#	Programming Topic	Mapped CO
	1	a) Overview of lab exercises, platform used etc. b) To determine the stated value of a resistor and inductor by interpreting the color code indicated on it. c) To get accustomed to the function of DC power supply, digital multimeter (DMM) and breadboard.	
1	2	a) To verify a given circuit using mesh analysis & observe practically b) To analyze the given circuit using nodal analysis & observe practically.	CO1
2	3	To verify the principle of Superposition theorem & find full load current for the given circuit practically.	CO1
3	4	a) To verify Thevenin's theorem and to find the full load current for the given circuit. b) To verify Norton's theorem and to find the full load current for the given circuit.	CO1
4	5	a) To analyze and obtain the time constant of a series RL circuit for a sinusoidal waveform. b) Also measure voltage across circuit elements and current flowing through it. c) Observe the phase relationship between applied voltage and current flowing through a series RL circuit.	CO2
5	6	a) To analyze and obtain the time constant of a series RC circuit for a sinusoidal waveform. b) Also measure voltage across circuit elements and current flowing through it. c) Observe the phase relationship between applied voltage and current flowing through series RC	CO2

		circuits.	
CO26	7	a) To Study the phenomenon of resonance in RLC series circuits. b) Determine the resonant frequency and bandwidth of the given circuit using a sinusoidal response. c) Plot the curve of current against the frequency (resonant curve).	CO2
7	8	To study and understand the construction of DC motor and identify the parts.	CO3
8	9	a) To study the characteristics of PN Junction diodes under forward and reverse bias conditions. Note the cut in voltage, static and dynamic resistance of diode. b) To study and verify Zener diode as a voltage regulator.	CO4
9	10	To construct a half wave rectifier using diode and observe its output waveform. To construct a single phase full-wave rectifier using diode and observe its output waveform.	CO4
10	11	Verification and interpretation of truth table for AND, OR, NOT, NAND, NOR, Ex-OR, Ex-NOR gates.	CO4

6. **Tutorial Plan-** Not applicable to BEEE

7. Course Material

Text Books (TB):

1. Ravish R Singh, “Basic Electrical Engineering”, McGraw Hill Education, 3rd Edition, 2019.
2. D. C. Kulshreshtha, “Basic Electrical Engineering”, McGraw Hill Education, 1st Edition 2017.
3. E. Hughes, “Electrical and Electronics Technology”, Pearson Education, 10th Edition 2013.
4. Boylstad R.L., Nashelsky L., “Electronic Devices and Circuit Theory”, Pearson, 12th Edition, 2012.
5. M. Morris Mano, “Digital Logic and Computer Design”, Prentice Hall India, 10th Edition 2008.

Reference Books:

1. V. D. Toro, “Electrical Engineering Fundamentals”, Pearson Education India, 2nd Edition 2015.
2. Jacob Millman & Halkias, "Electronic Devices & Circuits", Tata McGraw Hill, 2nd edition, 2013.

8. GenAI Usage

8.1 **Pre-class Activity:**

Technical notes on curriculum topics:

- Concept of electromagnetic Mutual Induction (for study of Transformers)
- P and N type semiconductors (for study of diodes, transistors and their applications)

8.2 **In-Class Activity:** Simple circuit design and analysis using GenAI

9. Academic Integrity Statement

Original work expected from students for all of the assigned assessment work. Copying in any form not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.